

Yixuan Liu

yixuan_liu1@brown.edu | (+86) 18669662539 | github.com/Ahhhh2016 | linkedin.com/in/yixuan-liu1/

Computer Graphics | Visual Computing | Rendering Engineering | AI Applications

Education

Brown University

M.S. in Computer Science, Visual Computing

Providence, RI, USA

Sep 2023 – Dec 2026 (Expected)

- Relevant Coursework: Computer Graphics, Computer Vision, Deep Learning, Artificial Intelligence

Southwest Jiaotong University

B.S. in Computer Science and Technology

Chengdu, China

Sep 2019 – Jul 2023

- GPA: 3.6/4.0; Second-Class Scholarship (top 5%); CCPC 2021 Bronze Medal (Women's Group); Sichuan Collegiate Programming Contest Silver Medal

University of Leeds

Sino-UK Joint Degree Program in Computer Science

Leeds, UK

Sep 2019 – Jul 2023

Technical Skills

Languages: C/C++, Python, JavaScript/TypeScript, Java, GLSL, SQL

Graphics: OpenGL, Qt, Eigen, Three.js, WebGL/GLSL, Ray/Path Tracing, BRDF, Monte Carlo Sampling, Shadow Mapping, Post-processing

Geometry/Simulation: Half-edge Mesh, Loop Subdivision, QEM, Isotropic Remeshing, Bilateral Denoising, BVH

Engineering: React, Vue3, Vite, Quarkus, Docker, Kubernetes, Redis, Git, CMake, OpenMP

Selected Graphics Projects

Physically-Based Monte Carlo Path Tracer

Brown CSCI 2240

C++17 / Eigen / Qt 6 / OpenMP / BVH / tinyobjloader

- Built a Monte Carlo path tracer from scratch, solving the rendering equation for soft shadows, color bleeding, caustics, mirrors, and dielectric refraction.
- Added next-event estimation, adaptive Russian Roulette, cosine-weighted / Phong-lobe importance sampling, BVH acceleration, and OpenMP row scheduling.

Half-Edge Mesh Geometry Processing Toolkit

Brown CSCI 2240

C++17 / Eigen / Qt 6 / Half-edge Mesh / tinyobjloader

- Implemented OBJ IO, Loop subdivision, QEM simplification, isotropic remeshing, and bilateral mesh denoising on a custom half-edge mesh.
- Used hash-indexed halfedges/edges for amortized $O(1)$ edge flip/split/collapse access; validated topology with 11 invariant checks.

Interactive ARAP Mesh Deformation

Brown CSCI 2240

C++17 / Eigen / Qt 6 / OpenGL / SimplicialLLT / SVD

- Implemented real-time ARAP anchor/drag deformation with SVD local rotations, sparse global solves, and cached Cholesky factorizations for interactive editing.

Real-Time FEM Soft-Body Simulation

Brown CSCI 2240

C++17 / Eigen / Qt 6 / OpenMP / Tetrahedral FEM

- Built a real-time tetrahedral soft-body simulator with StVK elasticity, damping, gravity, ground/sphere collisions, and mouse-driven manipulation.
- Computed deformation gradients, Green strain, and stresses per frame; advanced dynamics with midpoint integration and two-stage contact handling.

Tea for Two - Flight Edition

Real-Time Graphics Final Project

C++17 / OpenGL / GLSL / Qt / Perlin Noise / Shadow Mapping

- Built a first-person flight experience with procedural terrain/water, portal rendering, PCF shadows, and GLSL stylized effects including toon shading, outlines, fog, and motion blur.

Experience

Meituan

Beijing, China

Site Reliability Engineering Intern

Apr 2025 – Jun 2025

- Supported SRE operations through cross-region disaster-recovery research, alerting-script automation, and monitoring/log-analysis workflows.

UN Trade & Development – ASYCUDA

Geneva, Switzerland

Software Engineering Intern

May 2024 – Nov 2024

- Developed Quarkus backend services and reusable Vue 3 components; contributed to Docker/Kubernetes-based CI/CD delivery.
- Implemented JasperReports PDF export and migrated i18n mappings from JSON configuration to backend entity annotations.

Mengda AI

Remote

Co-founder / Full-stack Developer

Jul 2025 – Present

- Co-built an AI resume platform from 0 to 1, supporting editing, AI assistance, PDF export, JWT auth, analytics, and 200+ users.
- Built the editor UI with React/TypeScript/Vite/Tailwind; developed Flask + SQLAlchemy REST APIs and integrated Coze workflows.

Writing and Additional Projects

- Maintain a personal website and the Chinese tutorial series *Graphics After Tea*; additional work includes Stippling Studio, Monster Mash Reproduction, A Small Firework, and PomoKanban.